

The Economic Implications of Housing Supply

Glaeser and Gyourko (2018) || ECON 490

Taylor Mackay || Email: tmackay@fullerton.edu

Slides Overview

In these slides, we'll discuss:

1. How the paper thinks about the price of housing
2. Housing supply across the U.S.
3. Economic consequences of restricted supply
4. The political economy of housing reform

The Central Question

Why are some cities so expensive while others aren't?

- Glaeser and Gyourko's answer = ***regulation-constrained supply***

Paper provides:

- A way to measure how housing markets are performing
- An analysis of what happens when housing markets fail to perform well

Tobin's q for Housing

Intuitively, paper measures value of assets relative to replacement cost:

$$\mathbf{P/MPPC = Housing Price \div Minimum Profitable Production Cost}$$

- **MPPC** = what it actually costs to build a house (land, materials, labor, etc.)
- **P/MPPC > 1** suggests something is preventing new construction

If prices exceed costs but builders aren't building, something's wrong

What Goes Into MPPC?

Physical construction costs:

- *RS Means data*: roughly \$70–\$95/sq ft depending on market (economy-quality)
- Varies some by region, but not dramatically

Land costs:

- Vacant land sales rarely observed; imputed as 20% of (land + construction) via builder surveys

Normal profit margin:

- Builders need ~17% markup to stay in business

A Useful Benchmark

In a **competitive, unregulated market**, we'd expect:

$$P/MPPC \approx 1$$

Prices should hover around the cost of building new units

- If they exceed that, builders enter, supply increases, prices decline
- If $P/MPPC$ stays well above 1 for years, supply is constrained

What Do the Data Show?

National data from 2013 show ~74% of U.S. homes priced near or below MPPC

- These are "cheap" markets where building isn't constrained
- Some markets, however, have higher P/MPPC ratios

Metro Area	P/MPPC Ratio
San Francisco–Oakland–Hayward	2.84
LA–Long Beach–Anaheim	> 2
Oxnard–Thousand Oaks–Ventura	> 2

3 Types of Housing Markets

Glaeser and Gyourko define 3 types of housing markets

- We can map this classification to ranges of P/MPPC values

Market Type	Example	P/MPPC	What's Happening
Declining	Detroit	< 1	Prices below construction cost
Elastic	Atlanta	≈ 1	Supply responds to demand
Inelastic	San Francisco	$>> 1$	Regulation blocks supply

California as a Special Case

Coastal metros (especially California) dominate the high P/MPPC list

What constrains supply?

- **Zoning:** Density limits, single-family requirements
- **Local permitting:** Slow, expensive, uncertain
- **Community opposition:** NIMBYism

The authors call the gap between price and MPPC the **regulatory tax**

The Regulatory Tax

The regulatory tax is the implicit cost of all the barriers to building

For the median SF home (2013):

- MPPC = \$281,690 (construction + imputed land + 17% profit margin)
- Market price = \$800,000
- The "regulatory tax" $\approx$ \$518K
 - *Paper reports this as an implied land value of $\sim$ \$490K, $\sim 10\times$ the land value assumed in MPPC*

A Caveat

The authors note that the optimal tax on housing construction is likely **positive**

- Don't we want more housing?!
- In Orange County, yes, we should be building more

In general, however, construction imposes negative externalities

- Traffic disruption, noise, etc. during building
- Changes to built environment and service demand in longer run
- In principle, tax on building gets producers to internalize these costs

Paper argues regulatory tax is much larger than likely value of these negative externalities

Who Wins and Who Loses?

Winners:

- Existing homeowners (their assets appreciate)
- Current residents who value "neighborhood character"

Losers:

- Renters (pay higher rents) and would-be buyers (priced out)
- Workers who'd move to productive areas but can't afford to

This has distributional consequences

- Housing wealth becomes concentrated among those who already own

Labor Misallocation

High housing costs hurt the whole economy, not just individuals

- Productive cities have high wages *because* workers are productive there
- But workers can't afford to move there

What happens? Those workers decide to stay in less productive places

- Overall economic output is lower than it could be

Paper's back-of-the-envelope estimates find losses equal to ~0.7–2% of GDP

- Lower end implies ~\$200 billion per year in lost output (0.7% of ~\$31T GDP)

Barriers to Construction

If restricted supply is so costly, why not just allow more building?

- Political economy of housing reform is difficult to overcome
- Existing homeowners have **concentrated benefits** from high prices

Costs are diffuse – spread across renters, potential movers, the broader economy

- Homeowners vote; future residents don't
- Local governments have strong incentives to restrict supply

What Would Solutions Look Like?

1. State-level override of local zoning (e.g., California's SB 9, SB 35)
2. Streamlined environmental review for housing
3. By-right development that removes discretionary approval

CA under Governor Newsom has made progress here, but obstacles remain

Key distinction in housing policy: building housing is often both illegal ***and*** difficult

- Something like zoning reform means a builder is allowed to build
- But slow permitting can make projects prohibitively slow/difficult to complete